

Harsh Müller

Embedded Systems & IoT Firmware Engineer • 3 Years Experience • MERIQ Index: 74% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: PCB Debugging, I2C/SPI/UART, FreeRTOS, Python, ARM Cortex, MQTT, C/C++, Embedded Linux

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core PCB Debugging service, reducing latency by 34%.
- Led a squad of 5 engineers delivering scalable microservices processing 20M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 33%.

Associate Embedded Systems & IoT Firmware Engineer — *Bosch* (2023 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using FreeRTOS saving \$18k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [PCB Debugging, I2C/SPI/UART, FreeRTOS, Python]

Engineered end-to-end production solution leveraging PCB Debugging and I2C/SPI/UART with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [FreeRTOS, Python, ARM Cortex, MQTT]

Architected high-resilience system integrating FreeRTOS with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Software Engineering

Imperial College London — 2019 - 2023 (Score: 3.51 / 4.0)